

Kim Sung-Bin

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[Website](#) | [Scholar](#) | [Github](#)

RESEARCH INTEREST

Audio-visual learning, Multi-modal learning, Cross-modal generation, Multi-modal LLM

EDUCATION

Pohang University of Science and Technology, POSTECH <i>Ph.D. program, Electrical Engineering (Advisor: Prof. Tae-Hyun Oh)</i>	Mar. 2023 – Present Pohang, Korea
Pohang University of Science and Technology, POSTECH <i>Master of Electrical Engineering (Advisor: Prof. Tae-Hyun Oh)</i> Thesis title: Learning to Generate Visual Scene and Text Description from Sound	Mar. 2021 – Feb. 2023 Pohang, Korea
Handong Global University <i>Bachelor of Science, Major: Electrical Engineering</i>	Mar. 2015 – Aug. 2020 Pohang, Korea

EXPERIENCE

Research Intern <i>Adobe Research, Sound Design AI (SODA) group</i> Worked on video-to-audio generation and linearizing the audio codecs.	Jun. 2026 – Sep. 2026 San Francisco, California
Visiting Researcher <i>Speech, Audio, and Language Technologies Lab (SALT Lab) at UT Austin</i> Worked on automated video dubbing, advised by Dr. David Harwath.	Sep. 2024 – Mar. 2025 Austin, Texas
Student Leader <i>Algorithmic Machine Intelligence Lab (AMI Lab) at Dept. of EE, POSTECH</i> Served as the student leader of AMI Lab	Jun. 2023 – Jun. 2024 Pohang, Korea
Teaching Assistant <i>Dept. of EE, POSTECH</i> Serving as a teaching assistant on “Visual Intelligence” lecture.	Jan. 2021 – Dec. 2021 Pohang, Korea
NAVER & UPSTAGE BoostCamp Mentoring Program <i>NAVER & UPSTAGE</i> Mentoring on computer vision lectures.	Jan. 2021 – May. 2021 Pohang, Korea
Military Service <i>Vanguard Unit, Republic of Korea Army</i> Discharged as a sergeant.	Sep. 2016 – Jun. 2018 Cheorwon, Korea

AWARD & HONOR

CVPR 2026 Outstanding reviewer	Jun. 2026
Full-ride Scholarship during Ph.D. program <i>SBS Cultural Foundation (max \$75,000 for 4-years)</i>	Aug. 2023 – Present
Graduated SUMMA CUM LAUDE, Handong Global University	Aug. 2020
Best Student Paper Award, ICROS 2020	Jul. 2020
Full-ride Scholarship during B.S. program <i>National Science & Technology Scholarship (Total \$24,000)</i>	Mar. 2015 – Aug. 2020

PUBLICATION (INTERNATIONAL)

(Equal contribution are denoted by “*”.)

Abbreviations

SPL	<i>IEEE Signal Processing Letters</i>
TVCJ	<i>The Visual Computer Journal</i>
ICML	<i>Int. Conf. on Machine Learning</i>
ICLR	<i>International Conference on Learning Representation</i>
CVPR	<i>IEEE Conference on Computer Vision and Pattern Recognition</i>
ICCV	<i>IEEE Int. Conf. on Computer Vision</i>
MM	<i>ACM Int. Conf. on Multimedia</i>
AAAI	<i>AAAI Conference on Artificial Intelligence</i>
WACV	<i>IEEE Winter Conference on Applications of Computer Vision</i>
ICASSP	<i>IEEE International Conference on Acoustics, Speech and Signal Processing</i>
NAACL	<i>Annual Conference of the North American Chapter of the Association for Computational Linguistics</i>
ACL	<i>Annual Meeting of the Association for Computational Linguistics</i>
EMNLP	<i>Conference on Empirical Methods in Natural Language Processing</i>
TMLR	<i>Transactions on Machine Learning Research</i>
IJCV	<i>International Journal of Computer Vision</i>

Conference

- [J19] A paper on “Retrofitting Linearity into Pretrained Audio Codecs,” *Under-review*.
- [C18] **Kim Sung-Bin**, L. JungMok, J. Jung, O. Hyun-Bin, H. Ryu, D. Harwath, T.-H. Oh, “Hear, Localize, and Reason: Spatially Aware Scene Understanding for Audio-visual LLMs,” *Arxiv*, 2026. (*with a high peer-reviewed score*)
- [C17] **Kim Sung-Bin**, J. Jang, D. Harwath, T.-H. Oh, “FacEDiT: Unified Talking Face Editing and Generation via Facial Motion Infilling,” *Arxiv*, 2026. (*with a high peer-reviewed score*)
- [C16] O. Hyun-Bin, K. Shimada, Y. Takida, **Kim Sung-Bin**, T. Uesaka, T. Shibuya, K. Lee, T.-H. Oh, Y. Mitsufuji, “Spatio-Temporal Audio Language Modeling for Dynamic Sound Sources,” *EMNLP*, 2026.
- [C15] K. Kim, **Kim Sung-Bin**, O. Hyun-Bin, T.-H. Oh, “Physics-Aware Deepfake Detection via Distance–Speech Consistency,” *Interspeech*, 2026.
- [C14] S. Baek, L. JungMok, **Kim Sung-Bin**, T.-H. Oh, “Efficient Hyper-Parameter Search for LoRA via Language-aided Bayesian Optimization,” *ICML*, 2026.
- [C13] L. JungMok, **Kim Sung-Bin**, J. Jang, L. Hyun, T.-H. Oh, “SMILE-Next: Teaching Large Language Models to Detect, Classify, and Reason about Laughter,” *ACL*, 2026. (*Accepted as an Oral presentation*)
- [C12] J. Choi, J.-H. Kim, **Kim Sung-Bin**, T.-H. Oh, J. S. Chung, “AlignDiT: Multimodal Aligned Diffusion Transformer for Synchronized Speech Generation,” *MM*, 2025.
- [C11] **Kim Sung-Bin**, J. Choi, P. Peng, J. S. Chung, T.-H. Oh, D. Harwath, “VoiceCraft-Dub: Automated Video Dubbing with Neural Codec Language Models,” *ICCV*, 2025.
- [C10] L. Chae-Yeon*, O. Hyun-Bin*, H. Eungi, **Kim Sung-Bin**, S.-K Nam, T.-H. Oh, “Perceptually Accurate 3D Talking Head Generation: New Definitions, Speech-Mesh Representation, and Evaluation Metrics,” *CVPR*, 2025. (*Accepted as a highlight paper (3.7% acceptance rate)*).
- [C9] **Kim Sung-Bin***, O. Hyun-Bin*, J. Lee, A. Senocak, J. S. Chung, T.-H. Oh, “AVHBench: A Cross-Modal Hallucination Benchmark for Audio-Visual Large Language Models,” *ICLR*, 2025.
- [C8] **Kim Sung-Bin**, K. Jun-Seong, J. Ko, Y. Kim, T.-H. Oh, “SoundBrush: Sound as a Brush for Visual Scene Editing,” *AAAI*, 2025.
- [C7] **Kim Sung-Bin***, L. Chae-Yeon*, G. Son*, J. Ju, S.-K Nam, T.-H. Oh, “MultiTalk: Enhancing 3D Talking Head Generation Across Languages with Multilingual Video Dataset,” *Interspeech*, 2024.

- [C6] H. Eungi*, O. Hyun-Bin*, **Kim Sung-Bin**, C. N. Etcheberry, S.-K Nam, J. Ju, T.-H. Oh, “Enhancing Speech-Driven 3D Facial Animation with Audio-Visual Guidance from Lip Reading Expert,” *Interspeech*, 2024.
- [C5] L. Hyun*, **Kim Sung-Bin***, S. Han, Y. Yu, T.-H. Oh, “SMILE: Multimodal Dataset for Understanding Laughter in Video with Language Models,” *NAACL Findings*, 2024.
(Presented in Workshops on AV4D and MMFM, in conjunction with ICCV 2023)
- [C4] **Kim Sung-Bin**, L. Hyun, D.-H. Hong, S.-K Nam, J. Ju, T.-H. Oh, “LaughTalk: Expressive 3D Talking Head Generation with Laughter,” *WACV*, 2024.
- [C3] Kim Youwang, L. Hyun*, **K. Sung-Bin***, S.-K. Nam, J.-H. Joo, T.-H. Oh, “Spatio-Temporally Consistent Face Mesh Reconstruction on Videos,” *Workshop on 3DMV: Learning 3D with Multi-View Supervision (in conj. with CVPR)*, 2023.
- [C2] **Kim Sung-Bin**, A. Senocak, H. Ha, A. Owens, T.-H. Oh, “Sound to Visual Scene Generation by Audio-to-Visual Latent Alignment,” *CVPR*, 2023.
(Media coverage: covered by Yonhap News, SBS, and many local media. Also, presented in Workshop on AI4CC, in conjunction with CVPR 2023 and Workshop on AV4D, in conjunction with ICCV 2023. Invited paper talk in Workshop on Sound to Sight, in conjunction with CVPR 2023.)
- [C1] M.-K Kim*, **Kim Sung-Bin***, T.-H. Oh, “Prefix Tuning for Automated Audio Captioning,” *ICASSP*, 2023. (Selected as an Oral presentation. Media coverage: covered by Yonhap News, and SBS.)

Journal

* Journal impact factors; *TVCI*: **3.5**, *SPL*: **3.1**

- [J5] A paper on “Sound to image generation,” *IJCV under-review* (<https://arxiv.org/abs/2412.06209>).
- [J4] H. Ha, O. Hyun-Bin, K. Jun-Seong, K. Byung-Ki, **K. Sung-Bin***, L.-T Tran, J.-Y Kim, S.-H Bae, T.-H. Oh, “Revisiting Learning-based Video Motion Magnification for Real-time Processing,” *TMLR*, 2026
- [J3] K. Youwang, L. Hyun*, **K. Sung-Bin***, S.-K Nam, J.-H. Joo, T.-H. Oh, “A Large-Scale 3D Face Mesh Video Dataset via Neural Re-parameterized Optimization,” *TMLR*, 2024
- [J2] K. Byung-Ki, **Kim Sung-Bin**, T.-H. Oh, “The devil in the details: simple and effective optical flow synthetic data generation,” *The Visual Computer Journal (TVCI)*, Springer, 2024
- [J1] J.-E. Lee*, **Kim Sung-Bin***, S. Kang, T.-H. Oh, “Lightweight Speaker Recognition in Poincaré Spaces,” *Signal Processing Letters (SPL)*, 2021

PATENT

Image Generating Apparatus	P2023111-03-US
Motion Amplification Devices and Method of Using Thereof	Republic of Korea. 10-2021-0184021

ACADEMIC SERVICE

Journal Reviewer

- International Journal of Computer Vision (IJCV)
- IEEE Transactions on Audio, Speech and Language Processing (TASL)
- Transactions on Machine Learning Research (TMLR)

Conference Reviewer

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- International Conference on Computer Vision (ICCV)
- International Conference on Computer Vision (ECCV)
- International Conference on Machine Learning (ICML)

- International Conference on Learning Representations (ICLR)
- Conference on Neural Information Processing Systems (NeurIPS)
- Special Interest Group on Computer Graphics and Interactive Techniques (SIGGRAPH)
- British Machine Vision Conference (BMVC)
- ACM Multimedia (MM)

REFERENCE

Dr. Tae-Hyun Oh, Professor, KAIST

Relationship: M.S. & Ph.D. advisor

E-mail: thoh.kaist.ac.kr@gmail.com

Dr. Arda Senocak, Professor, UNIST

Relationship: Collaborator

E-mail: arda.senocak@gmail.com

Dr. David Harwath, Professor, UT Austin

Relationship: Advisor during visiting research period

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